

Dongyuan Zhai

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EDUCATION

Columbia University

Visiting Undergraduate at Department of Computer Science

Summer 2026

GPA: 4.16

The Hong Kong University of Science and Technology

June 2029

Bachelor of Engineering in Computer Science

GPA: 3.32

Xi'an Tie Yi High School

June 2025

STEM Student

Top 0.5% in Gaokao

COURSEWORK IN PROGRESS

Introduction to Artificial Intelligence

OOP and Data Structures in C++

Design and Analysis of Algorithms

Multivariable Calculus

COURSEWORK

Programming with C++

Data Structures in Java

Object-Oriented Programming in Java

Introduction to Python Programming

Linear Algebra

Discrete Mathematical Tools for CS

Calculus I

Calculus II

PROJECTS

Natural Language Processing Project: Math Word Problem Generation (Georgia Institute of Technology)

Summer 2026

- Used Git for project management and Conda environment for package management
- Used OpenAI GPT-5.6 Sol and Claude Fable 5 API calls as well as local Qwen3-8B and DeepSeek-R1-Distill-Llama-8B models to generate GSM8K style math word problems
- Used Matplotlib to plot math word problem generation accuracies

Interactive Entertainment Project: Classic Games of Tetris, Tank Wars and Lights Out (HKUST)

Fall 2025 – Spring 2026

- Implemented functions using Python to move a tetris block, change its shape and eliminate an entire row once built
- Implemented classes and functions using Python to drive and shoot a tank as well as trigger bomb explosions once shot
- Implemented GUI of Tetris and Tank Wars using Tkinter Modules
- Implemented functions using C++ to switch on and off lights in a matrix until all lights are turned off

SKILLS

Programming Languages: Python, C++, Java

Frameworks and Tools: NumPy, pandas, Matplotlib, Linux environment, Git

Languages: English (IELTS 7.5), Mandarin (Native), Cantonese